Paritosh Singh

Software Engineer — AI/ML & Full-Stack Development

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EDUCATION

SRM Institute of Science and Technology

B.Tech in Computer Science and Engineering (Data Science) - 8.69 June 2023 - May 2027

CERTIFICATIONS

Introduction to Machine Learning NPTEL (IIT Madras)

PROJECTS

Vigil – AI-Powered Multi-Hazard Disaster Prediction & Early Warning Platform 2026

- **Tech Stack:** Python, FastAPI, PostgreSQL, scikit-learn, XGBoost, LightGBM, SHAP, React, TypeScript
- Built a 5-hazard ML inference platform (flood, cyclone, drought, heatwave, compound) ingesting live data from NASA FIRMS, OpenWeatherMap, and Open-Meteo across 15 Indian regions; implemented SCBC, CWEF, and ARTC novelties for bias correction, ensemble fusion, and adaptive thresholding.
- Integrated SHAP explainability into prediction outputs; delivered a real-time dashboard with ROC-AUC aggregation and regional risk overlays via a FastAPI backend with 12+ route groups, JWT auth, and SQLAlchemy + PostgreSQL persistence.
- Designed and executed full model training pipelines for all 5 hazard types (Random Forest, XGBoost, LightGBM) with dedicated training scripts per model, a dataset admin orchestration workflow, and sanity-check and seeding scripts for operational validation across 15 monitored Indian regions.

MedTrack – Containerized Electronic Medical Record (EMR) System 2025

- **Tech Stack:** Docker, Docker Compose, Node.js, Express.js, PostgreSQL, React, RESTful APIs
- Orchestrated 4 isolated microservices via Docker Compose with zero-downtime restarts; engineered 20+ REST endpoints with JWT-based RBAC across 3 user roles covering patient records, scheduling, lab ingestion, and notifications.
- Implemented schema-per-tenant isolation in PostgreSQL preventing cross-clinic data leakage while supporting concurrent multi-clinic operations.
- Delivered 5 real-time analytical dashboards surfacing KPIs including patient intake, appointment fill rates, revenue metrics, and lab turnaround times — reducing manual reporting overhead by an estimated 60%.

mini-redis - TCP Server & In-Memory Store Built from Scratch in C++ 2026

- **Tech Stack**: C++, POSIX Sockets, RESP Protocol, Linux, Git
- Engineered the full TCP socket layer (bind, listen, accept, recv), a custom RESP wire protocol parser, and a command dispatcher handling PING, ECHO, SET, GET, and DEL — built entirely from scratch in C++.
- Built an in-memory key-value store using std::unordered_map with O(1) average-case lookup, implementing key expiry via std::chrono TTL tracking and a non-blocking epoll-based event loop supporting concurrent client connections.
- Demonstrates deep systems programming ability - direct syscall usage, buffer management, protocol parsing, and event-driven I/O architecture on Linux.

SKILLS

- **Languages:** Python, C, C++, SQL
- **AI/ML & Data:** Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, SHAP, EDA, Statistical Analysis, Bias Mitigation, Fairness Metrics, Matplotlib, Machine Learning, Data Visualization
- **Systems:** C++ Socket Programming, POSIX APIs, RESP Protocol, TCP/IP, Event-driven I/O
- **Backend:** FastAPI, Node.js, Express.js, RESTful API Design, JWT Auth, RBAC
- **Databases:** PostgreSQL, MySQL, Schema Design, Query Optimization
- **DevOps & Infra:** Docker, Docker Compose, Linux, Git
- **CS Fundamentals:** Data Structures & Algorithms, OOP, System Design, Security Principles

ACHIEVEMENTS

Winner (2nd Place) – Capture the Flag, SRM Institute of Science and Technology